



SAP HANA »

HANA Interview Questions

SAP HANA FAQ's

Which operating system is supported by HANA?

Currently SAP HANA supports SUSE Linux Enterprise Server x86-64 (SLES) 11 SP1

Operating System

What are some features of SAP HANA in-memory database?

SAP HANA in-memory database main features are:

- SAP HANA in-memory is a combination of row based, column based and Object Oriented base technology.
- It is Hybrid In-memory database.
- SAP HANA in-memory use parallel processing with multicore CPU Architecture.
- SAP HANA In-Memory database takes 5 nanoseconds to reads data whereas other conventional databases take 5 milliseconds to read memory data.

What is the Core Architecture of SAP HANA?

SAP HANA was first, developed in **Java** and C++ and was designed only to run on Suse Linux Enterprise Server 11 Operating System . SAP HANA system consists of multiple components which are responsible for enhancing

computing power of HANA system.

What are the Authentication methods in SAP HANA?

In order to get access to HANA database all SAP HANA users must be verified with different Authentications Mechanisms provided by SAP HANA. Various types of authentication method are supported by SAP HANA system. These authentication methods are configured while creating user profile.

Here is the list of authentication methods supported by SAP HANA:

- Kerberos
- SAML 2.0
- SAP Logon tickets
- X.509

Read here more **SAP HANA Authentication Mechanisms**

What are the different compression techniques in HANA?

Different types of compression techniques in

available in SAP HANA are:

- Run-length encoding
- Cluster encoding
- Dictionary encoding
- Prefix encoding
- Indirect encoding
- Sparse encoding

Read here more **Compression Types Exist in SAP HANA**

What is the development language used by SAP HANA?

C++ is the development language used by SAP HANA.

Read more about **C++ Programming Language**

Can user always upgrade SAP HANA directly, even if there is a significant difference between the source and the target level?

We can upgrade SAP HANA from any source level to any target level without an intermediate step. But there are some exceptions:

step. But there are some exceptions :

- Source level with SPS2 or below
- Exceptions mentioned explicitly in the related
(Example: SAP HANA SPS 08 Database
Revision 80)

What are the security features present in SAP HANA?

The following security related features are provided by SAP HANA :

- Authentication and SSO
- Authorization
- User and Role Management
- Encryption of data communication in Network
- Encryption of data in Persistence Layer

Can Intercompany be compatible with HANA?

Intercompany is certainly not compatible with HANA databases.

Dump DYNPRO_FIELD_CONVERSION occurs using transaction PA20/PA30 when accessing an entry of loans.

Please navigate transaction se16 table PA0045 and Pa0078 and checks for inconsistency and if there is any inconsistency remove it and error would be solved.

For avoiding this event in the future again control how It0078 is created and that no Z report modifies it.

How can the user create & export PlanViz in HANA Studio?

- Prepare the SQL Statement in the SQL editor. User right-click your mouse button and selects Visualize Plan.
- Clicking on the Save As icon and specifying the save location In the top right-hand corner of the screen.
- For viewing the executed plan, user clicks on the Execute dropdown in the right-hand corner of the screen and click on Execute (Display Results)
- Finally, clicking on the Save As icon in the right-hand corner and select your save location.

Note: User may be prompted to change to the SAP HANA PlanViz perspective. It is safe to click 'Yes' to this dialog box.

How to modify the SSFS master key in HANA System Replication?

Post altering the master key on the Primary system, user can perform the following steps for synchronizing the key on the Secondary site:

- Make sure that the SSFS_<SID>.KEY file is placed under
`/usr/sap/<SID>/SYS/global/hdb/security/ssfs/.`
- Now manually copy the master key file from the Primary system to the Secondary system.
- Delete the `ssfs_key_file_path` setting under [cryptography] in `global.ini` on both sites.
- Restart the systems to in order to make the changes effective.

Can the user consume HANA SQL logic in ABAP Perspective? In Other words, I want to write logic in HANA Studio (SQL console using HANA SQL and SQL script) and then utilize some method/

way for consuming it in ABAP perspective.

- In the ABAP Perspective, a SQL Console is present. User can copy/paste your HANA SQL statement and validate it there.
- If everything is accurate, then user creates an ABAP program and add "INTO TABLE @DATA(result)" to your statement.

List the steps for upgrading your HANA express edition to HANA 2?

The steps for upgrading HANA express edition to HANA 2 are listed below:

- First Increase memory to 8 GB
- Also, download the Binary Installer
- Please then create a temporarily user
- Uninstalling HANA 1 and Installing HANA 2
- At last, connect to HANA 2 Cockpit

How to a HANA cross join between two nodes in a Graphical Calculation View as HANA is allowing only left and right join?

A dummy column can be introduced into a

projection by returning a constant (for ex. 1) in both the nodes that user want to join. Now Join the dummy columns. Therefore, technically it is still an inner join but more effectively a cross join.

How to view User Detail in S/4 HANA Cloud?

- User Logs in as Administrator.
- Selecting App "Maintain Business Users".
- In the "Logon Status" Dropdown selects "All Users".
- Select "Go" for generating a list view of Users

How can the information regarding the HANA parameters, which can be configure online/offline?

- Open cdexe on the OS side, incase you already configured this alias else go to the folder "cd
`/usr/sap/$SAPSYSTEMNAME/SYS/exe"`
- Open the config folder by command "`cd config`"
- Open the Master ini files you need with the editor

- Check the hint for the parameter you are looking for, there you will find configure online or offline

How can the user enable Smart Data Access(SDA) trace?

Step 1) First please enable an SDA trace by running the with the following SQL in HANA Studio:

- alter system alter configuration ('indexserver.ini','SYSTEM') SET ('trace','fedtrace')='debug' with reconfigure;
- alter system alter configuration ('indexserver.ini','SYSTEM') SET ('trace','fedodbctrace')='debug' with reconfigure;
- alter system alter configuration ('indexserver.ini','SYSTEM') SET ('trace','federationexecution')='debug' with reconfigure;

Step 2) Now regenerate the problem

Step 3) Then please disable the fedtrace:

- alter system alter configuration ('indexserver.ini','SYSTEM') UNSET ('trace','fedtrace') with reconfigure;
- alter system alter configuration ('indexserver.ini','SYSTEM') UNSET ('trace','fedodbctrace') with reconfigure;
- alter system alter configuration ('indexserver.ini','SYSTEM') UNSET ('trace','federationexecution') with reconfigure;

Step 4) Now you can collect the indexserver trace files.

How can the user shrink SAP HANA Data volume size?

Use Syntax

```
ALTER SYSTEM RECLAIM DATAVOLUME [SPACE]  
[<host_port>]  
<percentage_of_payload_size> <shrink_mode>
```

This statement aims at reducing the data volume size to a percentage of payload size.

This statement works in a similar manner for defragmenting a hard drive. Scattered pages

around a data volume are moved to the front of the volume, and the free space at the end of the data volume is truncated.

How to Embed XSEngine in HANA?

- User adds a custom parameter named 'embedded.' to the httpserver section of the xsengine.ini file and setting its value to true.
- Resuming the HANA instance.

Note: that it is normal behaviour to see the XSEngine still appearing as a separate process in the process list at OS level

How can the CPU consumption of SAP HANA hosts be optimized?

The High system CPU consumption is caused by following process:

1.OS process

Solution: Optimize OS configuration

2.SAP HANA process

Solution: Optimize OS calls triggered by SAP

3.Other Processes

Solution: Optimize OS calls of other processes

4.Non SAP HANA processes

Solution: Optimize non SAP HANA software

5.High SAP HANA User

Solution: Analyze expensive SAP HANA operations

Is it possible to define a limit for the overall SAP HANA memory consumption?

The overall limit for the SAP HANA memory consumption is controlled via the global allocation limit:

```
global.ini -> [memorymanager] ->  
global_allocation_limit =  
<maximum_memory_allocation_in_MB>
```

The default value depends on the available physical memory and the SAP HANA revision level:

SPS 06 and below: 90 % of physical memory

SPS 07 and higher: 90 % of first 64 GB, 97 % of remaining physical memory

SAP HANA Network Interview Questions and Answers for Experienced

SAP HANA Network FAQ1. What is the meaning of network in the context of this SAP mean?"Network" in the context of this SAP means all communications between different processes regardless of

HANA Workload Management Interview Question and Answer

FAQ on SAP HANA Workload Management1.

What is SAP HANA workload management?

SAP HANA can be used for various types of workloads, such as: OLAP workload, for example reporting in BW systems OLTP workloa

SAP HANA Security Interview Question and Answer

FAQ: SAP HANA SecurityListed below are the SAP HANA alerts that indicate problems in the security area: Alert Name Description 57 Secure store fil

SAP HANA Database Server Management Console (hdbcons) Interview Question and answer

FAQ HANA DB Server Management Console (hdbcons) 1. What is the SAP HANA database server management console? The SAP HANA

server management console? The SAP HANA database server management console (hdbcons) is an expert tool delivered&

SAP HANA Admin Interview Question and Answers

SAP HANA Admin FAQs1. How do you perform BW-on-HANA Migration?Ans: DMO- Database Migration. This is a new feature available in SUM(Software Update Manager Tool).2. How many VMs are supported on a sing

HANA Expensive Statement Trace Interview Question and Answer

1. What purpose does expensive statements trace serve?To record execution details for database requests that exceed defined thresholds, the expensive statements trace is used. When doing SQL optimizat

SAP HANA Fast Data Access (FDA) Interview Questions and Answer

FAQ on SAP HANA Fast Data Access (FDA)

What is fast data access and what does it do?

Fast data access (FDA) is a protocol which is accessible in the environments of SAP ABAP. It allows for the submissi

SAP HANA LOBs Interview Question and Answer

HANA LOBs Question and Answer1. What do you mean by LOBs in SAP HANA

environments?To store a large amount of data such as text documents and images LOBs (large objects) are columns that are used.2. Wh

SAP HANA Real Time Interview Questions List

Here is a list of SAP HANA real time interview

questions which are frequently asked:1)

Explain what is SAP HANA?2) Explain the two types of Relational Data stored in HANA?3)

Explain what is modeling s

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SAP HANA View Cache Interview Questions & Answers

FAQs: SAP HANA View Cache1. What does view cache allow?To cache results on SQL views, calculation views and CDS views the SAP HANA view cache (or result cache

SAP HANA view cache (or result cache extension) is used. This allows quick dat

SAP HANA Statement Routing Interview Questions for Experienced

What is the procedure of implementation of statement routing?The following picture shows how statement routing is implemented. During preparation table information is collected and sent back to the cl

IBM and Lenovo SAP Hana Support Script Interview Q&A

FAQ: SAP HANA Support ScriptSAP HANA Memory LimitsQuery:If left unconfigured, each installed and running HANA instance may use up to 97% (90% in older HANA revisions) of the system's memory. If

SAP HANA Database Trace Interview

Questions and Answer

SAP HANA Database Trace FAQ'sQ-1 What is the structure of HANA database trace?Ans)

The trace entries have the following structure:

```
[<thread_id>]{<connection_id>}
```

```
[<transaction_id>/
```

SAP HANA Studio Interview Questions and Answers

FAQ: SAP HANA Studio 1. Define SAP HANA Studio? The SAP HANA Studio is both the main administration tool and the central development environment for SAP HANA. 2. Can individual SQL statements

HANA Smart Data Access & BW Near-Line Storage Interview Q&A

What are the requirement of using SAP HANA Smart Data Access with BW Near-Line Storage? SAP-NLS Solution with SAP IQ SAP BW 7.40 SP5 or higher SAP HANA as Database, SP6 or higher SAP IQ 16.0

SAP HANA Database Backup & Recovery Interview Question and Answer

FAQ SAP HANA Database Backup & Recovery Where is the detailed information related to SAP HANA backup and recovery available? You can find further details about backup and recovery in the SAP HANA&nb

HANA Technical Performance Optimization Service (TPO) Interview Q&A

FAQ on SAP HANA Technical Performance

Optimization Service (TPO)What is the SAP HANA Technical Performance Service (TPO)?

The SAP HANA Technical Performance Optimization (TPO) is an expert service whic

Loads and Unloads in SAP HANA Environments Interview Questions and Answers

What are loads and unloads in SAP HANA environments? When a column is loaded, it is copied from persistence into SAP HANA column store memory.Unloads are table column displacements from column store m

SAP HANA Cockpit Interview Questions and Answer

FAQ SAP HANA CockpitWhat does SAP HANA Cockpit 1.0 refer to?The SAP HANA cockpit 1.0 basically is a SAP Fiori Launchpad site which provides you with a single point of access to a

provides you with a single point-of-access to a range of Web-based ap

HANA Services and Ports Interview Questions and Answer

FAQ SAP HANA Services and PortsWhat are services and ports in SAP HANA?As part of the SAP HANA instance, all the services running in the context of this SAP document are processes mentioned on operati

HANA Runtime Dump Interview Questions and Answers

HANA Runtime Dump FAQ'sQ-1) Define SAP HANA runtime dump?Ans) SAP HANA Dump is a text file which provides information about SAP HANA database current status. These runtime dumps helps SAP or SAP u

SAP HANA SQL Optimization Interview Questions and Answer

FAQ SAP HANA SQL Optimization Here are some top most important and frequently asked SAP HANA SQL Optimization Interview Questions and Answer such as What is an expensive SQL statement,

SAP HANA Data Aging Interview Questions and Answer

FAQ HANA Data Aging in SAP S/4HANAAre there any fundamentals for using Data Aging in SAP S/4HANA?Data Aging is mainly available in SAP Business Suite on HANA and S/4HANA Applications (starting with SA

SAP HANA Backup Encryption Interview Question and Answer

SAP HANA Backup Encryption FAQWhat does SAP HANA backup encryption refer to?Backup encryption denotes that authorized parties can access the backups which have been stored outside a secure area.How do

Multitenant Database Container (MDC) Interview Questions and Answers

Multitenant Database Container (MDC)

FAQs What do you mean by SAP HANA

Multitenant Database Containers? The SAP

HANA system can be now installed in multiple-container mode, therefore it can contain mult

Column Store and Row Store Interview Questions & Answers

Column Store and Row Store FAQ's Q) Can a table be assigned in both column and row store? Ans) No, only one store can be assigned to a table at any time which can be either column store or r

HANA Inverted Individual Indexes Interview Questions and Answers

SAP HANA Inverted Individual Indexes FAQs

1. Explain inverted individual indexes? An inverted individual index is referred to as a special type of SAP HANA index specifically for column store ta

SAP HANA Execution Engine (HEX) Interview Questions and Answers

SAP HANA Execution Engine (HEX) FAQs How to control usage of HEX User can use the following parameter on global level in order to activate or deactivate HEX: indexserver.ini -> [sql] ->

FF7AN/FF7BN Interview Questions and Answers

T-code FF7AN/FF7BN FAQs Why there is no data is displayed by FF7AN/FF7BN or cash position details app? FF7AN or cash position details app display the data from FQM_FLOW table. Please check this

SAP HANA Sequences Interview Questions & Answers

SAP HANA Sequences FAQ's 1) Explains sequences in SAP HANA environments? A sequence is a database object in SAP HANA environments which can be used to generate unique numbers. 2) What are

SAP HANA Extended SQL Executor (ESX) Interview Questions and Answers

SAP HANA Extended SQL Executor (ESX) FAQs 1) Define SAP HANA Extended SQL Executor (ESX)? SAP HANA Extended SQL Executor (ESX) is a front-end execution engine which replaced the row engine. it r

SAP HANA Performance Optimization FAQs

Which are the prerequisites should be fulfilled for performing a detailed performance

analysis? The performance analysis capabilities significantly improve with the latest SAP HANA patch level

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